ORIGINAL PAPER

DENTAL ANXIETY BEFORE AND AFTER ENDODONTIC MICROSURGERY

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INTRODUCTION

Endodontic microsurgery is a procedure that a tooth could require after the failure of previous treatments [1]. Consequently, the patient may suffer from a high level of concern, stress, fear, or anxiety. It has been researched that dental anxiety can affect a variety of aspects of life, including physiological, behavioral, and health [2]. Therefore, it is necessary to measure dental anxiety to avoid its impacts. When dentists try to infer the presence of dental anxiety in their patients, they often think it is lower than it really is [3]. Therefore, dentists need scales to clearly identify dental anxiety. There are several scales and ways to measure dental anxiety; some of the most used scales are Corah's dental anxiety scale, modified dental anxiety scale, dental anxiety inventory, etc. Those scales and other are mostly verbal rating scales; hence, difficulties in the interpretation of terms

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could arise [4]. On the other hand, given that the visual analogue scale (VAS) is a non-verbal rating scale, it is not connected with any particular descriptor of the case study [5]. Besides, it is simple and quick, and provides a method to evaluate anxiety before and after a particular circumstance of stress [6].

OBJECTIVES

The aim of this study was to measure and compare dental anxiety before and after endodontic microsurgery using VAS.

MATERIAL AND METHODS

This longitudinal study was conducted at the dental clinic of Universidad Peruana Cayetano Heredia, Lima, Perú. The study protocol was approved by the Ethics Committee of Universidad Peruana Cayetano Heredia (file No. 215-10-19).

PARTICIPANTS

The participants were previously scheduled to undergo endodontic microsurgery. Between 2011 and 2017, in average, microsurgery was used to treat 35 patients per year in the dental clinic. The selection of the sample was non-probabilistic. Given that the mean number of patients in previous years was not so large, we decided to use the same number for our sample. Exclusion criteria involved any psychiatric or neurological disease reported in the medical record and participants aged under 18. Forty-one participants were gathered in eight months, six of which were excluded for not having filled out the scales completely. A total of 35 participants agreed and signed a written informed consent.

MEASUREMENTS

The visual analogue scale (VAS, 0-10 cm) was used to assess dental anxiety. It was applied before and after endodontic microsurgery. The rating scale ranged from "not anxious at all" to "worst experience ever". The distance between "not anxious at all" and the patient-made mark was calculated and provided a quantitative variable that was used for statistical evaluation [7]. This kind of scale was employed in similar studies. For statistical purposes and to classify variables into categories, cut-off scores presented in a previous study was used [8], in which score 3 represented participants who experienced anxiety.

PROCEDURE

The participants completed VAS while they were waiting for their endodontic microsurgery in the waiting room of the dental clinic. Microsurgery was performed among 2nd year postgraduate students of endodontics, and a clinical supervisor monitored the procedure. The microsurgery lasted on average 2-3 hours. The second VAS was completed after microsurgery, while patients were waiting for their medical prescription. The same researcher conducted all the assessments.

CONCLUSIONS

Within the limits of the present study, we did not find a statistical difference between pre- and post-endodontic microsurgery anxiety levels.

CONFLICT OF INTEREST

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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